

SOFTWARE ENGINEERING LAB – 2 (UE24CS341A)

Agile Backlog Creation & Sprint Simulation in Jira

Team Number: 001

Problem Statement 1: Smart Lab Equipment Slot Reservation Portal

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EPIC: Epic 1 – Check Slot Availability (UC-05)

Note: This report covers only Epic 1, which comes from use case UC-05 (Check Slot Availability) in the Lab 1 report. The other use cases are covered separately by my teammates.

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Lab Objective

This lab converts use case UC-05 and requirements FR-001 and FR-003 from the Lab 1 report into an Agile Epic and User Stories in Jira. It then estimates effort using the Fibonacci scale, prioritises the backlog, simulates two one-week sprints on a Scrum board, and tracks sprint progress using the Burndown Chart.

1. Epic (Derived from UC-05: Check Slot Availability)

The use case and its requirements are grouped into one Epic. FR-001 requires students to view real-time availability before reserving a slot. FR-003 requires the system to prevent two students from reserving the same equipment for overlapping time slots, which depends on checking existing reservations first.

Epic Key	Epic Name	Mapped Requirements	Description
T0-1	Epic 1: Check Slot Availability	UC-05; FR-001, FR-003	Enable students to check whether equipment is free for a chosen date and time slot, and see alternative slots if it is already reserved.

2. User Stories for Epic 1

The Epic is broken into 6 User Stories using the format: As a [role], I want [goal], so that [benefit].

Key	Story	As a...	I want...	So that...
T0-2	1.1 Select Equipment and Time Slot	Student	to select a calibrated piece of equipment and a date and time slot	I can request the exact slot I need
T0-3	1.2 Verify Calibration Before Confirming	Student	the system to check the equipment's calibration status before confirming my reservation	I only book equipment that is fit for use
T0-4	1.3 Prevent Overlapping Reservations	Student	the system to reject a request that overlaps an existing reservation of the same equipment	my booking is never double-booked
T0-5	1.4 Enforce 2-Hour Limit and 7-Day Window	Student	the system to limit a reservation to 2 hours and to dates within the next 7 days	reservations stay fair and manageable for all students
T0-6	1.5 Lock Slot and Generate Reservation Token	Student	the slot to be locked and a reservation token generated when my booking is confirmed	I have proof of my booking
T0-7	1.6 Show Reservation Confirmation	Student	to see a confirmation of my reservation with the equipment, time and token	I know the booking succeeded

Acceptance criteria were taken from Lab 1. For example, FR-001: the student can view real-time availability of equipment. FR-003: when two requests overlap, only one reservation is confirmed and the other is rejected or redirected to an available slot. The use case flow also requires that when a slot is unavailable, the system displays alternative available slots.

3. Jira Backlog – Epic, User Stories and Story Points

The screenshot shows Epic 1 with all six User Stories, the priority icon and story points on each row, and the Epic panel reporting 6 work items and an estimate of 19 points.

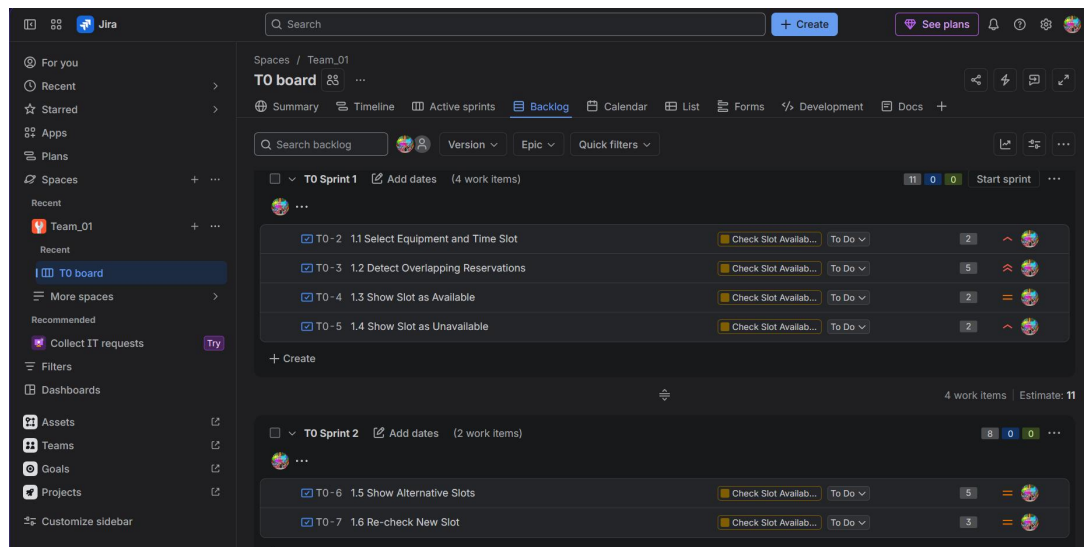


Figure 1: Jira Backlog – Epic 1 (T0-1) with stories T0-2 to T0-7 (2, 5, 2, 2, 5, 3 pts = 19 pts).

4. Backlog Prioritization

User Stories were ordered by business value and dependency. The backlog order in Jira is 1.2, 1.1, 1.3, 1.4, 1.5, 1.6.

Key	Story	Priority	Story Points
T0-3	1.2 Detect Overlapping Reservations	Highest	5
T0-2	1.1 Select Equipment and Time Slot	High	2
T0-4	1.3 Show Slot as Available	Medium	2
T0-5	1.4 Show Slot as Unavailable	High	2
T0-6	1.5 Show Alternative Slots	Medium	5
T0-7	1.6 Re-check New Slot	Medium	3

- **1.2 (Highest):** detecting overlaps is the core of the use case and directly supports the double-booking requirement (FR-003).
- **1.1, 1.3 and 1.4 (High):** selecting a slot and showing the Available or Unavailable result are the main success and alternate outcomes of the flow, so Sprint 1 delivers a complete basic check.

- **1.5 and 1.6 (Medium):** alternative slots and re-checking are helpful extensions that only make sense once the basic check works, so they are placed in Sprint 2.

Total Story Points (Epic 1): 19 | Highest: 5 | High: 4 | Medium: 10

5. Story Point Estimation – Planning Poker

Story points were estimated using the Fibonacci scale (1, 2, 3, 5, 8, 13...) based on complexity, effort and uncertainty, and the team agreed on the final estimates.

Story ID	Story Title	Lakshanyaa	Arfan	Kshama	Manjunatha	Agreed
T0-2 / 1.1	Select Equipment and Time Slot	2	2	3	2	2
T0-3 / 1.2	Detect Overlapping Reservations	5	5	8	5	5
T0-4 / 1.3	Show Slot as Available	2	2	2	3	2
T0-5 / 1.4	Show Slot as Unavailable	2	2	3	2	2
T0-6 / 1.5	Show Alternative Slots	5	5	8	5	5
T0-7 / 1.6	Re-check New Slot	3	3	2	3	3

Where estimates differed (stories 1.1, 1.2, 1.3, 1.4, 1.5 and 1.6), the members explained their reasoning and the team settled on the majority value.

6. Sprint Planning & Simulation

Sprint 1 (Duration: 1 week)

Sprint Goal: Students can select a slot and see whether it is available or unavailable.

Key	Story	Priority	Pts	Status
T0-2	1.1 Select Equipment and Time Slot	High	2	Done
T0-3	1.2 Detect Overlapping Reservations	Highest	5	Done
T0-4	1.3 Show Slot as Available	High	2	Done
T0-5	1.4 Show Slot as Unavailable	Medium	2	Done

Sprint 1 Velocity: 11 pts committed | 11 pts completed | 0 pts rolled over

Sprint 2 (Duration: 1 week)

Sprint Goal: Students can see alternative slots when a slot is taken and re-check the slot they choose.

Key	Story	Priority	Pts	Done
T0-6	1.5 Show Alternative Slots	Medium	5	Done
T0-7	1.6 Re-check New Slot	Medium	3	Done

Sprint 2 Velocity: 8 pts committed | 8 pts completed | 0 pts rolled over

Jira Sprint Board (Active Sprint view) – Sprint 1

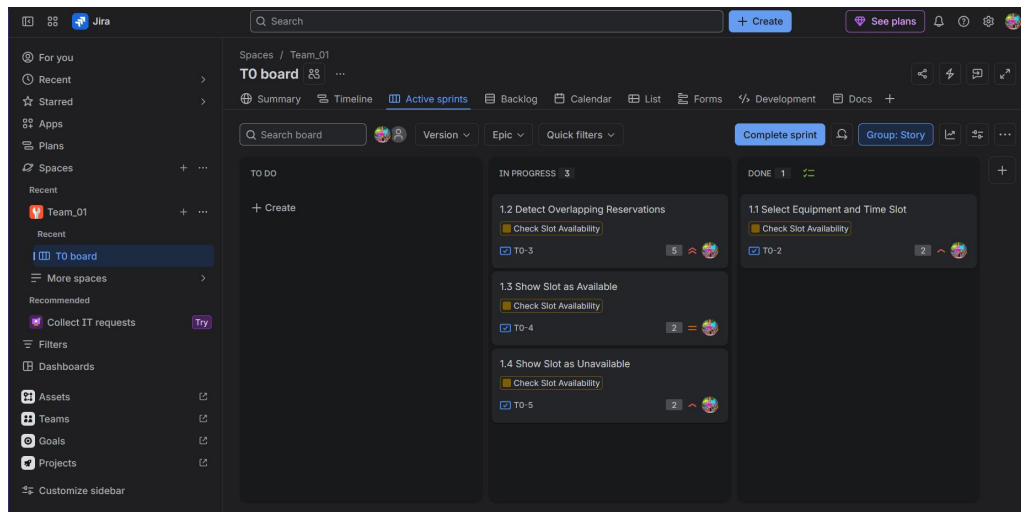


Figure 2: Sprint 1 Active Sprint board mid-sprint – Story 1.1 in DONE; the other stories IN PROGRESS

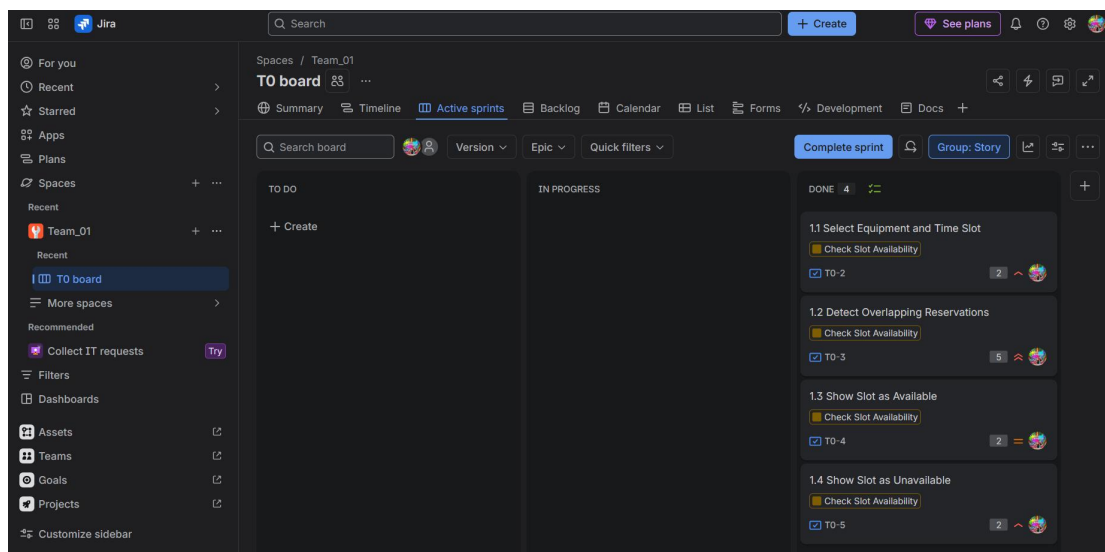


Figure 3: Sprint 1 Active Sprint board – all four stories (12 pts) in DONE, before clicking Complete sprint.

Sprint 2 outcome

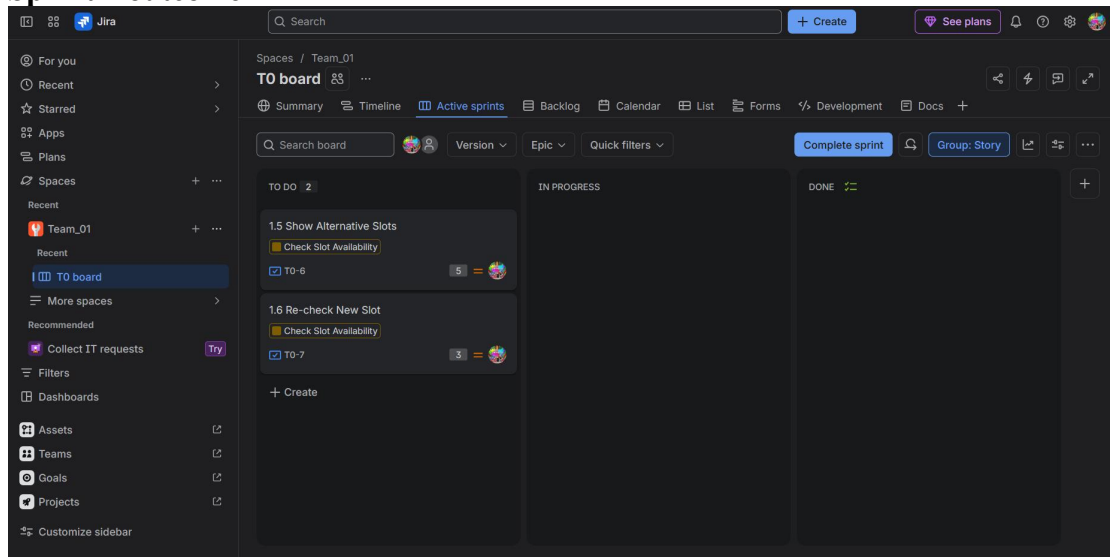


Figure 4: Sprint 2 Active Sprint board – both stories (8 pts) in TO DO.

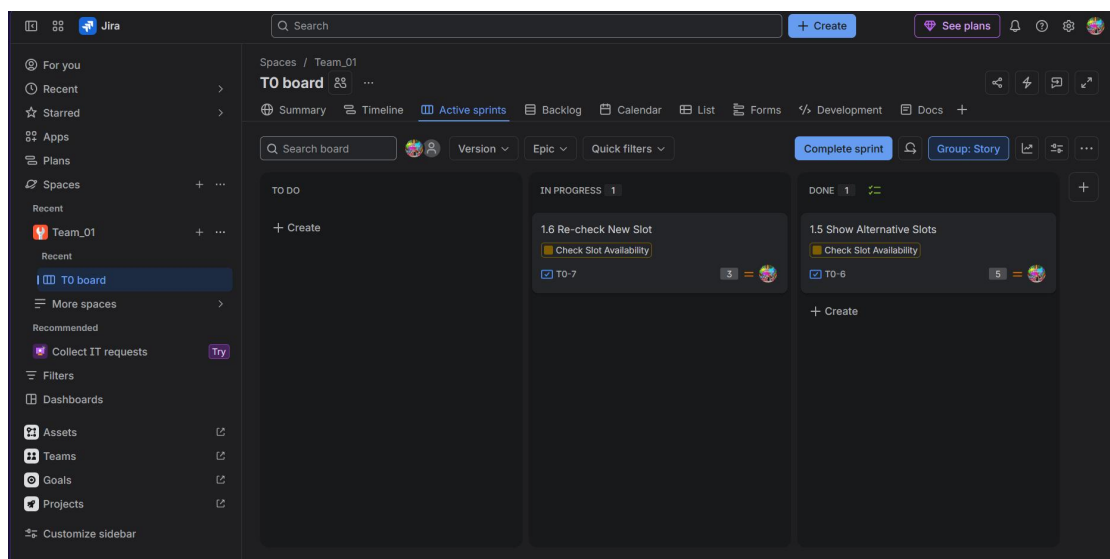


Figure 5: Sprint 2 Active Sprint board mid-sprint – Story 1.5 in DONE; Story 1.6 IN PROGRESS.

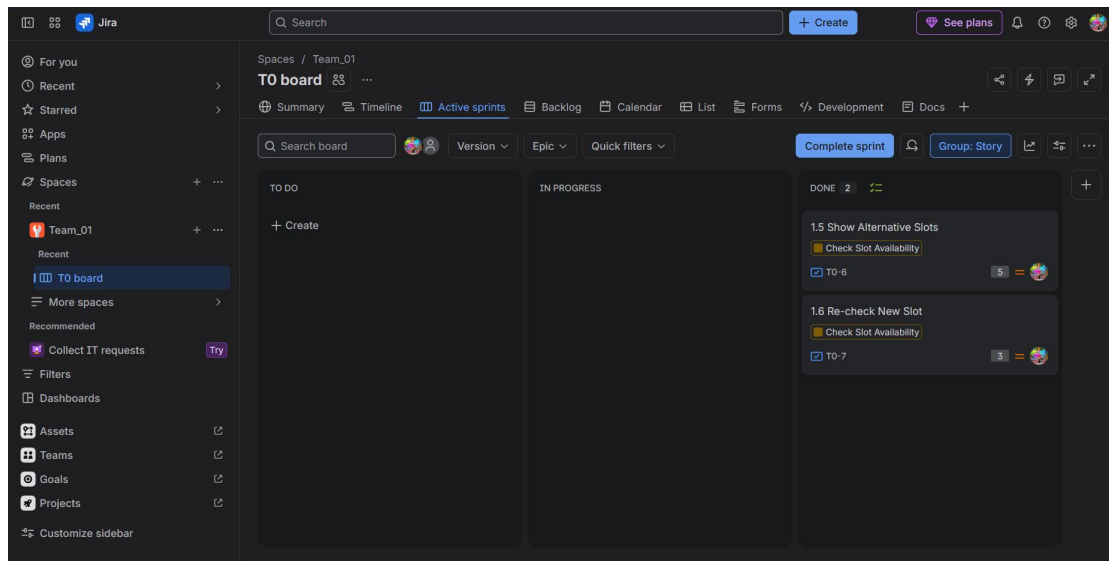


Figure 6: Sprint 2 Active Sprint board – both stories (8 pts) in DONE, before clicking Complete sprint.

7. Burndown Chart Analysis

The Jira Burndown Chart plots story points remaining (red line) against the ideal guideline (grey line). Each sprint is shown twice: once with non-working days displayed (hatched) and once with them hidden, which makes the guideline easier to read. The event table under each chart lists the exact completion times.

Sprint 1 Burndown (11 pts)

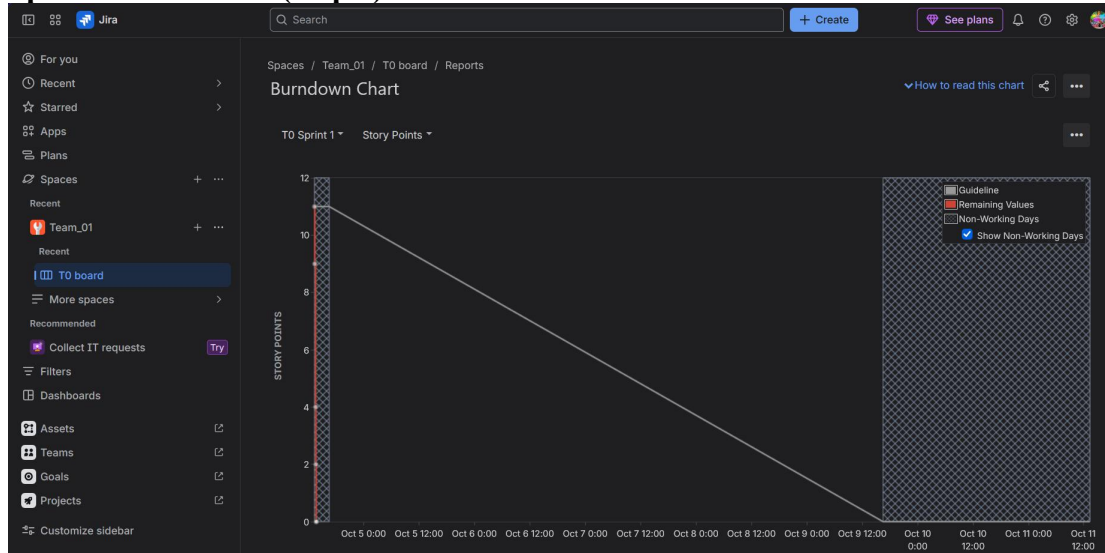


Figure 8: Sprint 1 Burndown, non-working days shown. $11 \rightarrow 9 \rightarrow 4 \rightarrow 2 \rightarrow 0$ between 1:22 PM and 1:39 PM on 4 Oct.

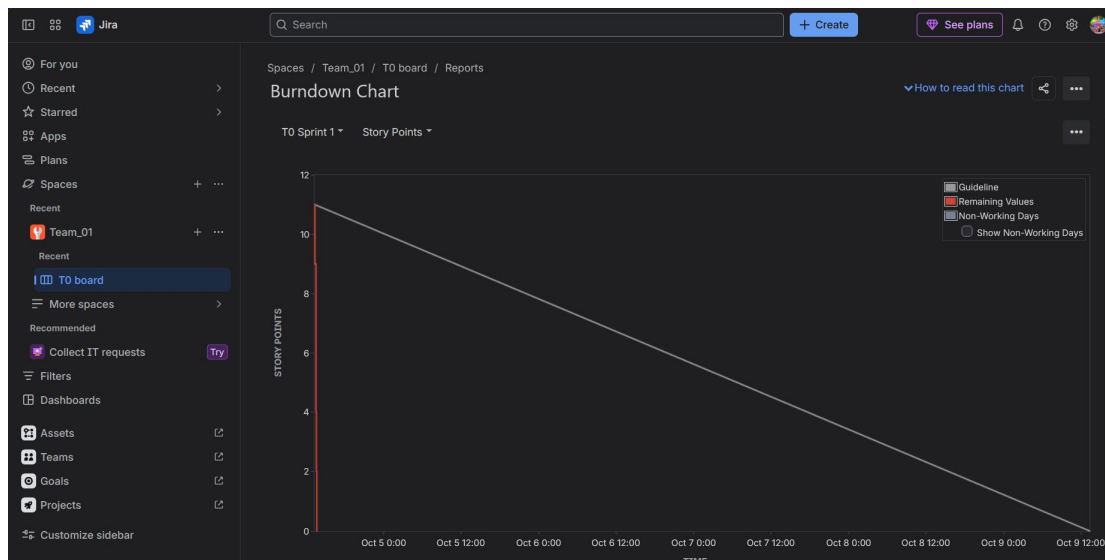


Figure 9: Sprint 1 Burndown, non-working days hidden. The sprint was ended at 1:40 PM, so the whole sprint took 22 minutes.

Sprint 2 Burndown (8 pts)

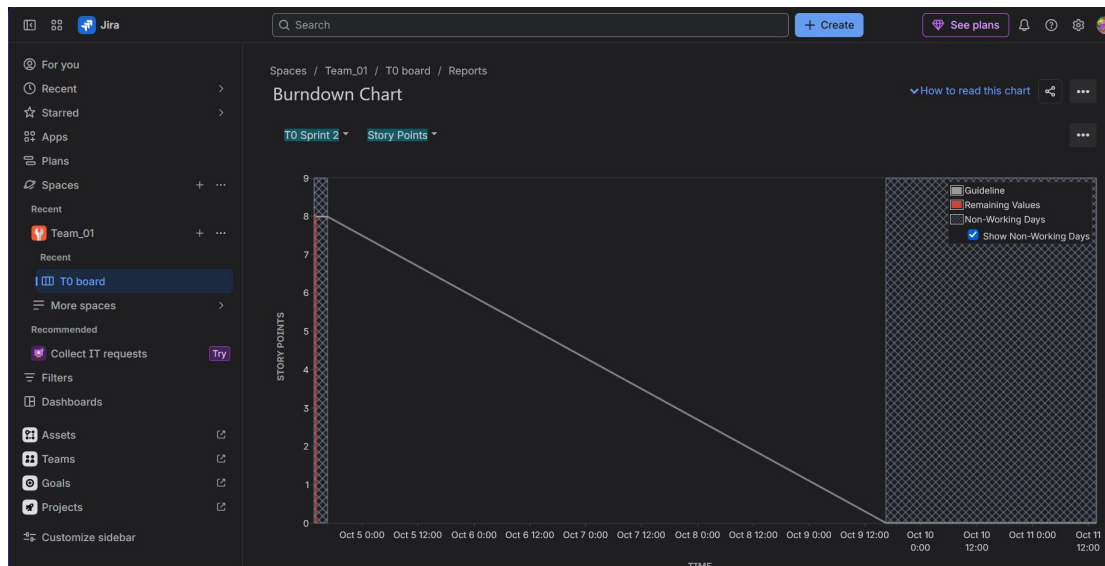


Figure 10: Sprint 2 Burndown, non-working days shown. $8 \rightarrow 3 \rightarrow 0$ between 1:57 PM and 2:02 PM on 4 Oct.

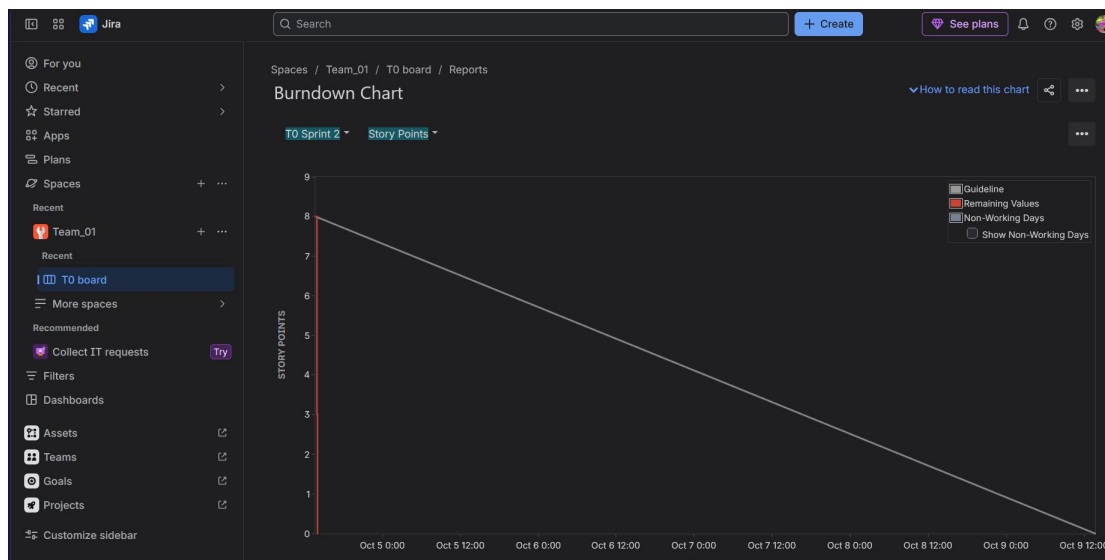


Figure 11: Sprint 2 Burndown, non-working days hidden. The sprint was ended at 2:03 PM, so the whole sprint took 22 minutes.

Summary of the burndown data (from the Jira event tables)

Sprint	Start (points)	Completion events	End (points)
Sprint 1	1:18 PM – 11	1:22 PM T0-2 (-2) $\rightarrow$ 9; 1:32 PM T0-3 (-5) $\rightarrow$ 4; 1:35 PM T0-4 (-2) $\rightarrow$ 2; 1:39 PM T0-5 (-2) $\rightarrow$ 0	0
Sprint 2	1:41 PM – 8	1:57 PM T0-6 (-5) $\rightarrow$ 3; 2:02 PM T0-7 (-3) $\rightarrow$ 0	0

8. Reflection Questions

1. Did your estimations reflect the actual effort?

We can only partly judge this. The sprints were simulated by moving cards on the board, so no real development effort was spent and the estimates cannot be checked against actual hours. What the exercise did show is that the relative sizing was sensible. Story 1.2 (overlap detection) and story 1.5 (alternative slots) received the highest estimate (5) because they involve comparing time ranges against existing reservations and searching for free slots, which carry the most complexity and uncertainty, while the display stories 1.3 and 1.4 and the selection story 1.1 were sized at 2. In a real sprint we would expect 1.2 and 1.5 to be the most likely to overrun. The largest disagreements in Planning Poker were on stories 1.2 and 1.5, where one member voted 8 instead of 5 because overlap detection and searching for alternative slots involve edge cases such as back-to-back and partially overlapping bookings. After discussion the team agreed on 5, since the logic is a time-range comparison rather than a new subsystem. This discussion helped align the estimates with the relative complexity and uncertainty of each story.

2. Was your backlog well-prioritized?

Yes. Story 1.2 was ranked Highest because detecting overlapping reservations is the core of this use case and supports the requirement to prevent double-booking. Stories 1.1, 1.3 and 1.4 were placed in Sprint 1 with it, so that the main success flow and the "slot unavailable" outcome could be delivered together. Stories 1.5 and 1.6 (alternative slots and re-checking) extend the basic check, so they were placed in Sprint 2. One possible improvement is that 1.4 and 1.5 could be planned together, since the alternative slots are shown at the moment a slot is found to be unavailable.

3. How did your simulated sprint align with your plan?

Both sprints were completed as planned. Sprint 1 committed and completed 11 points and Sprint 2 committed and completed 8 points, a total of 19 points with nothing rolled over, and both sprint goals were met. The board moved stories from To Do to In Progress to Done, and each story was finished before the sprint was closed. Because the work was simulated, Sprint 1 took about 22 minutes (1:18 PM to 1:40 PM) and Sprint 2 about 22 minutes (1:41 PM to 2:03 PM) instead of the one week that was planned, so the 100% completion reflects the simulation, not real delivery capacity.

4. What insights did the burndown chart give about your team's capacity?

In both sprints the red line dropped to zero within about 20 minutes of the start, far below the grey guideline, which is why it appears as a thin line at the left edge of the chart. In a real sprint that pattern would indicate that the team had under-committed. The event tables still gave useful information. Sprint 1 started with 11 points at 1:18 PM and burned down to zero by 1:39 PM, with the 5-point story 1.2 alone accounting for almost half of the burndown in one step. Sprint 2 started with 8 points at 1:41 PM and reached zero by 2:02 PM, with the 5-point story 1.5 accounting for most of its burndown. A realistic velocity would be somewhere around 8 to 11 points per week, but this would need to be re-measured with real work. The charts also suggest that large stories should be split so that progress is visible throughout the sprint.